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## TP5 BDA: Transactions et contrôle de concurrence

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ING2 INFORMATIQUE  
Sup Galilée

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Tout le TP a été réalisé avec une base de données MySQL

## Commande à lancer avant chaque début de session

```
SET AUTOCOMMIT = 0;
```

### Exercice n° 1

#### 1. Création de la table

```
MariaDB [tp5]> CREATE TABLE transaction ( idTransaction VARCHAR(44), valTransaction INT(10) );  
Query OK, 0 rows affected (0,013 sec)
```

#### 2.

La table est vide

```
MariaDB [(none)]> SET AUTOCOMMIT = 0;  
Query OK, 0 rows affected (0,000 sec)  
  
MariaDB [(none)]> USE tp5;  
Database changed  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T1',1);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T2',2);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T3',3);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> UPDATE t  
tee test tp5  
MariaDB [tp5]> UPDATE transaction SET valTransaction = 33 WHERE idTransaction = 3;  
ERROR 1292 (22007): Truncated incorrect DECIMAL value: 'T1'  
MariaDB [tp5]> UPDATE transaction SET valTransaction = 33 WHERE idTransaction = 'T3';  
Query OK, 1 row affected (0,001 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
MariaDB [tp5]> DELETE FROM transaction WHERE idTransaction = 'T1';  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> RO  
ROLLBACK ROUND ROW ROW_FORMAT  
ROLLUP ROUTINE ROW_COUNT ROWS  
MariaDB [tp5]> ROLLBACK ;  
Query OK, 0 rows affected (0,003 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)
```

#### 3.

Les données insérées sont perdues

```
ing@ing: ~  
MariaDB [tp5]> CREATE TABLE transaction ( idTransaction VARCHAR(44), valTransaction INT(10) );  
Query OK, 0 rows affected (0,013 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T4',4);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T5',5);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> ROLLBACK;  
Query OK, 0 rows affected (0,003 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]>  
  
ing@ing: ~  
MariaDB [tp5]> ROLLBACK;  
ROLLBACK ROUND ROW ROW_FORMAT  
ROLLUP ROUTINE ROW_COUNT ROWS  
MariaDB [tp5]> ROLLBACK ;  
Query OK, 0 rows affected (0,003 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T1',1);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T2',2);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T3',3);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> quit;  
Bye  
ing@ing: $
```

4.

Aucune donnée n'a été préservée

```
ing@ing: ~  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T5',5);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> ROLLBACK;  
Query OK, 0 rows affected (0,003 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T4',4);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]> INSERT INTO transaction VALUES ('T5',5);  
Query OK, 1 row affected (0,001 sec)  
  
MariaDB [tp5]>
```

```
MariaDB [tp5]> SELECT * FROM transaction;  
Empty set (0,001 sec)  
  
MariaDB [tp5]>
```

5.

Les données insérées sont présentes lorsque une colonne a été rajoutée et que le SGBD a validé les opérations précédentes

```
MariaDB [tp5]> SELECT * FROM transaction;
Empty set (0,001 sec)

MariaDB [tp5]> INSERT INTO transaction VALUES ('T3',3);
Query OK, 1 row affected (0,001 sec)

MariaDB [tp5]> INSERT INTO transaction VALUES ('T2',2);
Query OK, 1 row affected (0,001 sec)

MariaDB [tp5]> ALTER TABLE transaction ADD val2transaction INT(10);
Query OK, 0 rows affected (0,010 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [tp5]> ROLLBACK;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5]> SELECT * FROM transaction;
+-----+-----+-----+
| idTransaction | valTransaction | val2transaction |
+-----+-----+-----+
| T3           | 3             | NULL           |
| T2           | 2             | NULL           |
+-----+-----+-----+
2 rows in set (0,001 sec)
```

6.

- Une **transaction** est une série d'opérations de lecture et d'écriture effectuées sur la base de données.
- Une **session** représente un programme ou une connexion qui interagit avec la base de données.
- Une **transaction** est validée par un COMMIT.
- Une **transaction** peut être annulée par un ROLLBACK.

## Exercice n° 2

Après avoir créer le schéma de la base on insère quelques données :

```
MariaDB [tp5_ex2]> INSERT INTO vol VALUES ('V1',10,0);
Query OK, 1 row affected (0,007 sec)

MariaDB [tp5_ex2]> □
```

```

MariaDB [tp5_ex2]> insert into client VALUES ('C1','Client1',0);
Query OK, 1 row affected (0,007 sec)

MariaDB [tp5_ex2]> insert into client VALUES ('C2','Client2',0);
Query OK, 1 row affected (0,007 sec)

MariaDB [tp5_ex2]> 

```

## Isolation des transactions

```

MariaDB [tp5_ex2]> SET AUTOCOMMIT = 0;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5_ex2]> UPDATE vol SET nbrPlacesReservesVol = 1 WHERE idVol = 'V1';
Query OK, 1 row affected (0,008 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> UPDATE client SET nbrPlacesReservesClient = 1 WHERE idClient = 'C1';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 1                    |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
ERROR 1146 (42502): Table 'tp5_ex2.client' doesn't exist
MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 1                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

```

MariaDB [tp5_ex2]> SET AUTOCOMMIT = 0;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 0                    |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 0                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

## COMMIT et ROLLBACK

```

MariaDB [tp5_ex2]> ROLLBACK;
Query OK, 0 rows affected (0,007 sec)

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 0                    |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 0                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

```

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 0                    |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 0                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

```

MariaDB [tp5_ex2]> COMMIT;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 0                    |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 0                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

```

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReservesVol |
+-----+-----+-----+
| V1    | 10         | 0                    |
+-----+-----+-----+
1 row in set (0,002 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReservesClient |
+-----+-----+-----+
| C1      | Client1     | 0                      |
| C2      | Client2     | 0                      |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]> 

```

## Isolation incomplète = incohérence possible

Dans cette partie, on va mettre le **niveau d'isolation** à READ COMMITTED pour reproduire le problème des mises à jour perdues. Voici la commande à exécuter au début de la session :

```
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
```

Voici la série de commandes réalisée :

```
-- Transaction 1 : Consultation
SELECT * FROM vol WHERE idVol = 'V1';
SELECT * FROM client WHERE idClient = 'C1';
```

```
-- Transaction 2 : Consultation
SELECT * FROM vol WHERE idVol = 'V1';
SELECT * FROM client WHERE idClient = 'C2';
```

```
-- Transaction 1 : Mise à jour
UPDATE vol SET nbrPlacesReserveesVol = 2 WHERE idVol = 'V1';
UPDATE client SET nbrPlacesReserveesClient = 2 WHERE idClient = 'C1';
```

```
-- Transaction 2 : Mise à jour
UPDATE vol SET nbrPlacesReserveesVol = 3 WHERE idVol = 'V1';
UPDATE client SET nbrPlacesReserveesClient = 3 WHERE idClient = 'C2';
```

```
MariaDB [tp5_ex2]> UPDATE vol SET nbrPlacesReserveesVol = 2 WHERE idVol = 'V1';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> UPDATE client SET nbrPlacesReserveesClient = 2 WHERE idClient = 'C1';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> COMMIT;
Query OK, 0 rows affected (0,003 sec)

MariaDB [tp5_ex2]> SELECT * FROM vol WHERE idVol = 'V1';
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReserveesVol |
+-----+-----+-----+
| V1    | 10          | 3                     |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReserveesClient |
+-----+-----+-----+
| C1      | client1      | 2                         |
| C2      | client2      | 3                         |
+-----+-----+-----+
2 rows in set (0,001 sec)

MariaDB [tp5_ex2]>
MariaDB [tp5_ex2]>
MariaDB [tp5_ex2]>
MariaDB [tp5_ex2]>
MariaDB [tp5_ex2]>

MariaDB [tp5_ex2]> UPDATE vol SET nbrPlacesReserveesVol = 3 WHERE idVol = 'V1';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> UPDATE client SET nbrPlacesReserveesClient = 3 WHERE idClient = 'C2';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [tp5_ex2]> COMMIT;
Query OK, 0 rows affected (0,003 sec)

MariaDB [tp5_ex2]> SELECT * FROM vol;
+-----+-----+-----+
| idVol | capaciteVol | nbrPlacesReserveesVol |
+-----+-----+-----+
| V1    | 10          | 3                     |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client WHERE idClient = 'C2';
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReserveesClient |
+-----+-----+-----+
| C2      | client2      | 3                         |
+-----+-----+-----+
1 row in set (0,001 sec)

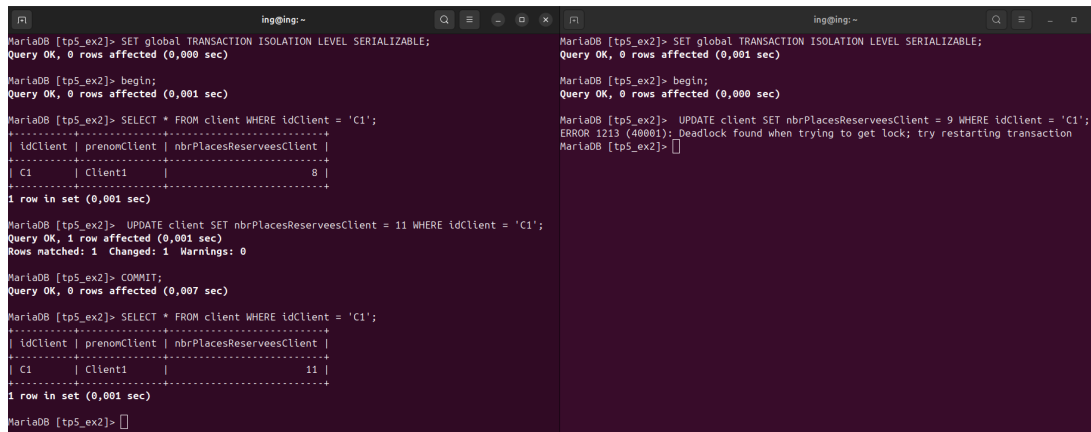
MariaDB [tp5_ex2]> SELECT * FROM client;
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReserveesClient |
+-----+-----+-----+
| C1      | client1      | 2                         |
| C2      | client2      | 3                         |
+-----+-----+-----+
```

On remarque qu'un des deux mises à jour a été perdue.

## Isolation complète = blocage et rejet des transactions possibles

Voici la série de commandes réalisée :

```
S1 : SET global TRANSACTION ISOLATION LEVEL SERIALIZABLE;
S2 : SET global TRANSACTION ISOLATION LEVEL SERIALIZABLE;
S1 : begin;
S2 : begin;
S1 : SELECT * FROM client WHERE idClient = 'C1';
S2 : UPDATE client SET nbrPlacesReserveesClient = 9 WHERE idClient = 'C1'; -- Bloqué
S1 : UPDATE client SET nbrPlacesReserveesClient = 11 WHERE idClient = 'C1';
S2 : Transaction rejetée suite à un deadlock
S1 : COMMIT;
```



```
MariaDB [tp5_ex2]> SET global TRANSACTION ISOLATION LEVEL SERIALIZABLE;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5_ex2]> begin;
Query OK, 0 rows affected (0,001 sec)

MariaDB [tp5_ex2]> SELECT * FROM client WHERE idClient = 'C1';
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReserveesClient |
+-----+-----+-----+
| C1      | client1      | 8                        |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]> UPDATE client SET nbrPlacesReserveesClient = 11 WHERE idClient = 'C1';
Query OK, 1 row affected (0,001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [tp5_ex2]> COMMIT;
Query OK, 0 rows affected (0,007 sec)

MariaDB [tp5_ex2]> SELECT * FROM client WHERE idClient = 'C1';
+-----+-----+-----+
| idClient | prenomClient | nbrPlacesReserveesClient |
+-----+-----+-----+
| C1      | client1      | 11                       |
+-----+-----+-----+
1 row in set (0,001 sec)

MariaDB [tp5_ex2]>

MariaDB [tp5_ex2]> SET global TRANSACTION ISOLATION LEVEL SERIALIZABLE;
Query OK, 0 rows affected (0,001 sec)

MariaDB [tp5_ex2]> begin;
Query OK, 0 rows affected (0,000 sec)

MariaDB [tp5_ex2]> UPDATE client SET nbrPlacesReserveesClient = 9 WHERE idClient = 'C1';
ERROR 1213 (40001): Deadlock found when trying to get lock; try restarting transaction
MariaDB [tp5_ex2]>
```

## Conclusion

MariaDB comme ORACLE utilise bien l'algorithme de verrouillage à 2 phases.

La seule différence majeure entre **MariaDB** et **Oracle** réside dans le **niveau d'isolation par défaut** :

- Pour **MariaDB** : REPEATABLE READ
- Pour **Oracle** : READ COMMITTED